Bihar Engineering University, Patna

B.Tech 5th Semester Examination, 2025

Course: B.Tech
Code: 101507

Subject: Transportation Engineering

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
*(iv) Question No. **1** is compulsory.*

Q.1 **Choose the correct option/answer the following *(Any seven question only)*:** **[2 x 7 = 14]**

(a) The approximate length of national highway in India is
(i) 1000 km (iii) 5000 km
(ii) 10000 km (iv) 50000 km

(b) The most accessible road is
(i) National highway (iii) Major District Road
(ii) State highway (iv) Village Road

(c) The Star and grid pattern of road network was adopted in
(i) Nagpur Road plan (iii) Bombay road plan
(ii) Lucknow road plan (iv) Delhi road plan

(d) A vehicle was stopped in 2 secs by fully jamming the beaks. The skid marks measured 9.8 meters. The average skid resistance coefficient will be
(i) 0.7 (iii) 0.4
(ii) 0.5 (iv) 0.25

(e) Full amount of super-elevation on a horizontal curve is provided at the
(i) Beginning of transition curve (iii) End of transition curve
(ii) Centre of the circular curve (iv) Centre of transition curve

(f) Assuming longitudinal coefficient of friction to be 0.4, the resulting retardation of a vehicle being brought to stop is nearly
(i) 0.98 m/s² (iii) 2.90 m/s²
(ii) 1.95 m/s² (iv) 3.93 m/s²

(g) If the difference in elevation between the edges of a pavement of width 9.0 m and its crown is 7.5 cm, what is the camber of the pavement?
(i) 1 in 60 (iii) 1 in 30
(ii) 1 in 45 (iv) 1 in 15

(h) When two roads with two-lane, two-way traffic, cross at an uncontrolled intersection, the total number of potential major conflict points would be
(i) 32 (iii) 16
(ii) 24 (iv) 4

(i) Which one of the following equipment is useful in determining spot speed in traffic engineering?
(i) Enoscope (iii) Radar
(ii) Periscope (iv) Tachometer

(j) Which one of the following is recommended for wet and cold climate?
(i) 80/100 penetration asphalt (iii) Cutback
(ii) Tar (iv) Emulsion

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Q.2 **(a)** What are the different modes of transportation? Explain the specific function of each of them. **[7]**

(b) Explain the classification of roads based on location and function as suggested in the Nagpur Road plan. **[7]**

Q.3 **(a)** Define 'Stopping Sight Distance'. Derive an expression for SSD for a road section where the design speed is V kmph and the coefficient of longitudinal friction is f. **[7]**

(b) Explain the importance of friction offered by road surface. Also discuss about the factors which influence highway friction.

Q.4 **(a)** Calculate the minimum sight distance required to avoid a head on collision of two cars approaching from the opposite direction at 100 kmph and 80 kmph on a road section. Assume a reaction time of 2.5 seconds, coefficient of friction of 0.7 and brake efficiency of 50% in either case. **[7]**

(b) What factors influence the design of vertical curves? Explain. **[7]**

Q.5 **(a)** What are the objectives of speed studies? What are the methods of presentation of speed data? **[7]**

(b) Explain the design procedure of Traffic signals according to Webster method. **[7]**

Q.6 Design a new flexible pavement for a two-lane undivided carriageway using the following data: Design CBR value of subgrade = 8.0%, Initial traffic on completion of construction = 1800cv per day, Average growth rate = 6.0% per year, Design life = 15 years, VDF value = 2.5. **[14]**

Q.7 **(a)** What are the various tests carried out on bitumen? Briefly mention the principle and uses of each Test. **[7]**

(b) List different tests on road aggregates and mention their advantages and limitations. **[7]**

Q.8 **(a)** The radius of a horizontal circular curve is 100m. The design speed is 50 kmph and the design Coefficient of lateral friction is 0.15. Calculate the superelevation required if full lateral friction is assumed to develop. **[7]**

(b) Classify different types of joints in CC pavements and mention the needs of each. **[7]**

Q.9 ***Write Short notes on any two of the following:*** **[7 x 2 = 14]**

(a) Mention step by step procedure of Marshall method of bituminous mix design.

(b) Different types of transition curves commonly adopted.

(c) Describe briefly about PIEV theory.

(d) Difference between flexible and rigid pavements.

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